

Confidence Intervals Step-by-Step

MDragon Data Tools - Study Guide

What is a Confidence Interval?

A **confidence interval** provides a range of plausible values for a population parameter, with a specified level of confidence.

Interpretation

"We are 95% confident that the true population mean is between 50 and 60."

Correct: The method captures the true parameter 95% of the time.

Incorrect: "95% probability the parameter is in this interval" (it either is or isn't).

CI for Population Mean (σ known)

Formula

$$CI = \bar{x} \pm z^* \times (\sigma / \sqrt{n})$$

where z^* = critical value for desired confidence level

Common Critical Values

Confidence Level	z^*
90%	1.645
95%	1.96
98%	2.33
99%	2.58

Example 1

 **Problem:** Sample of 50 has $\bar{x}=100$, $\sigma=15$. Find 95% CI.

 **Solution:**

$$\begin{aligned} \text{CI} &= 100 \pm 1.96 \times (15/\sqrt{50}) \\ &= 100 \pm 1.96 \times 2.12 \\ &= 100 \pm 4.16 \\ &= (95.84, 104.16) \end{aligned}$$

✓ **Answer:** 95% CI: (95.84, 104.16)

CI for Population Mean (σ unknown)

Formula

$$\text{CI} = \bar{x} \pm t^* \times (s / \sqrt{n})$$

where t^* depends on df and confidence level

Example 2

 **Problem:** Sample of 15 has $\bar{x}=72$, $s=8$. Find 90% CI.

 **Solution:**

$$\begin{aligned}df &= 14 \\t^* &= 1.761 \text{ (from t-table)} \\CI &= 72 \pm 1.761 \times (8/\sqrt{15}) \\&= 72 \pm 1.761 \times 2.066 \\&= 72 \pm 3.64 \\&= (68.36, 75.64)\end{aligned}$$

✓ **Answer:** 90% CI: (68.36, 75.64)

CI for Population Proportion

Formula

$$CI = \hat{p} \pm z^* \times \sqrt{[\hat{p}(1-\hat{p})/n]}$$

where $\hat{p} = x/n$ (sample proportion)

Example 3

 **Problem:** 120 out of 200 support a policy. Find 95% CI for true proportion.

 **Solution:**

$$\begin{aligned}\hat{p} &= 120/200 = 0.60 \\CI &= 0.60 \pm 1.96 \times \sqrt{[0.60(0.40)/200]} \\&= 0.60 \pm 1.96 \times 0.0346 \\&= 0.60 \pm 0.068 \\&= (0.532, 0.668)\end{aligned}$$

✓ Answer: 95% CI: (53.2%, 66.8%)

Sample Size Determination

For Mean

$$n = (z^* \sigma / ME)^2$$

where ME = desired margin of error

Example 4

 **Problem:** How many samples needed for 95% CI with ME=2, $\sigma=10$?

 **Solution:**

$$\begin{aligned} n &= (1.96 \times 10 / 2)^2 \\ &= (9.8)^2 \\ &= 96.04 \\ \text{Round up: } n &= 97 \end{aligned}$$

For Proportion

$$n = \hat{p}(1-\hat{p})(z^*/ME)^2$$

Use $\hat{p}=0.5$ if no estimate available (most conservative)

Factors Affecting CI Width

1. **Sample Size (n):** Larger n → Narrower CI
2. **Confidence Level:** Higher confidence → Wider CI

3. Standard Deviation: Larger σ or $s \rightarrow$ Wider CI

Trade-offs

- Want narrow CI? Increase sample size or decrease confidence
 - Want high confidence? Accept wider CI or increase sample size
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Practice with our calculators to build intuition!

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